

Detection of Invasive Ductal Carcinoma (IDC)

Final Year Project Proposal
Bachelors of Computer Sciences (BSCS)
By

S#	Name	Registration # / Roll # / Section	Mobile #	E-Mail
1.	Muhammad Uzair Ishfaq	SP-2020/BSCS/030/A	+92 3154784468	Sp20-bscs-030-@lgu.edu.pk
2.	Khadija Ibrahim	SP-2020/BSCS/011/A	+92 312 4301367	Sp20-bscs-011-@lgu.edu.pk

Supervised by:
Miss Gulshan Saleem

_____ (Signature)



Department of Computer Science
Lahore Garrison University
Lahore

Abstract

Breast cancer is a significant health concern worldwide, with invasive ductal carcinoma (IDC) being the most common subtype. In 2020, there were 2.3 million women diagnosed with breast cancer and 685 000 deaths globally [1]. Early and accurate detection of IDC is crucial for successful treatment and improved patient outcomes. Histopathological examination of breast tissue samples is a standard diagnostic procedure for breast cancer. However, manual interpretation of histopathology images can be time-consuming and subject to inter-observer variability. Also mostly doctors use their experience to examine these histopathology images under microscope which can lead to wrong diagnoses and loss of a life, Therefore, there is a need for an automated and accurate method for the detection of IDC. In this proposal, we present a project aimed at developing an AI-based system for the detection of IDC in breast histopathology images.

Introduction

Breast cancer is a significant health concern worldwide, with invasive ductal carcinoma (IDC) being the most common subtype. The standard diagnostic procedure for breast cancer is histopathological examination of breast tissue samples obtained through biopsy or surgery. The manual interpretation of histopathology images is time-consuming, labor-intensive, and subject to inter-observer variability, making it prone to errors. Therefore, there is a need for an automated and accurate method for the detection of IDC in breast histopathology images. Artificial intelligence (AI) [2] has shown great potential in improving the accuracy and efficiency of cancer diagnosis. In this project proposal, we propose the development of an AI-based system for the detection of IDC in breast histopathology images.

Problem statement

1. Manual interpretation of histopathology images for the detection of IDC is prone to errors.
2. Due to late detection of IDC patient can reach to such stage where it's become difficult to save his/her life.
3. Mostly doctors examine these histopathology images based on their experience which can lead to human error and wrong detection of IDC.

The manual interpretation of histopathology images for the detection of IDC is prone to errors and subject to inter-observer variability, leading to inaccurate diagnosis and delayed treatment.

Therefore, there is a need for an automated and accurate method for the detection of IDC in breast histopathology images. With this automated system we can easily and accurately identify IDC with in a patient which will help doctors to start their treatment as early, reduce the rate of mortality and wrong diagnosis.

Literature review

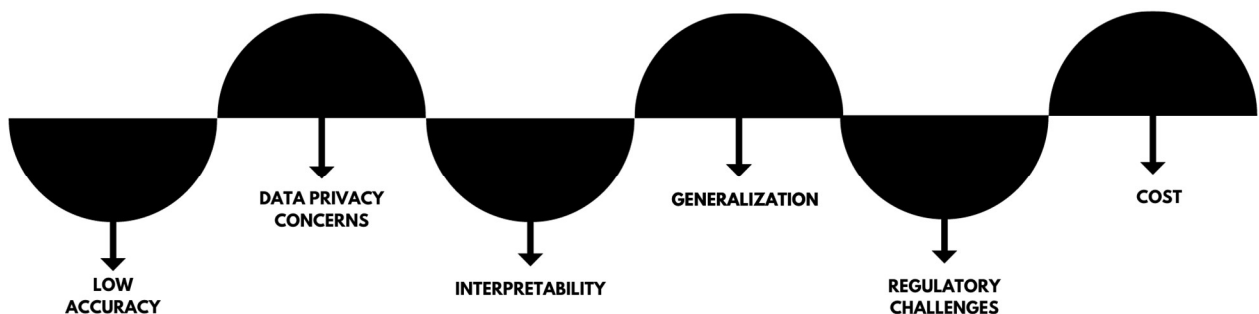
Numerous studies have made significant strides in utilizing AI-based systems for breast cancer detection in histopathology images. Among these, Convolutional Neural Networks (CNNs),

Transfer learning, A Deep Analysis of Transfer Learning Based Breast Cancer Detection Using Histopathology Images [3], Deep learning applications to breast cancer detection by magnetic resonance imaging [4] and many more have become a prevailing choice due to their efficiency in image classification tasks.

However, existing AI-based solutions in the market do have some limitations.

These include:

1. **Low accuracy:** All these AI models doesn't have good accuracy which leads to false prediction or classification.
2. **Data Privacy Concerns:** Many solutions require access to sensitive patient data, raising concerns about data privacy and security.
3. **Interpretability:** Deep learning models often lack transparency in their decision-making processes, making it challenging for healthcare professionals to understand why a particular diagnosis was made.
4. **Generalization:** Some AI models may not generalize well to diverse patient populations or imaging conditions, potentially leading to biased results.
5. **Regulatory Challenges:** Regulatory approval for medical AI solutions can be a lengthy and complex process, limiting their availability in certain regions.
6. **Cost:** Implementation and maintenance of AI systems can be costly, which may limit their adoption in resource-constrained healthcare facilities.



In our Project, we will use state-of-the-art literature and the latest advancements in deep learning to contribute to the ongoing efforts in improving breast cancer detection. By addressing the drawbacks of existing solutions and building upon the latest, we aim to create a more accurate, transparent, and accessible AI-based system for breast cancer diagnosis.

Project scope

Overview/Goal of the Project:

The primary goal of this project is to develop an innovative AI-based solution for the accurate and efficient detection of Invasive Ductal Carcinoma (IDC) in breast histopathology images. Breast cancer is a global health concern, and early detection is crucial for improved patient outcomes.

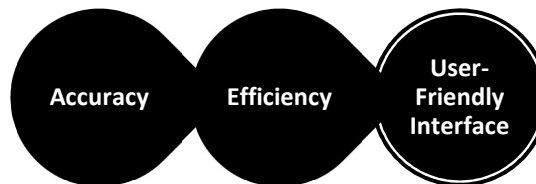
Manual interpretation of these images is time-consuming and prone to errors, necessitating the need for an automated system.

Proposed Solution:

Our solution will leverage cutting-edge and state of the art deep learning techniques, to analyze breast histopathology images and identify IDC with a high level of accuracy.

The proposed system will set itself apart from others by:

- 1. Accuracy:** We will employ state-of-the-art deep learning models and a carefully curated dataset to achieve high accuracy in IDC detection. Our goal is to surpass the current standards of accuracy in breast cancer diagnosis.
- 2. Efficiency:** The system will be designed to provide results swiftly, aiding medical professionals in making timely decisions for patient treatment.
- 3. User-Friendly Interface:** The final output will be a user-friendly web application, ensuring accessibility for medical practitioners. It will be designed with a focus on ease of use and interpretation of results.



Differentiation from State-of-the-Art:

Our proposed solution aims to surpass the state-of-the-art in several ways:

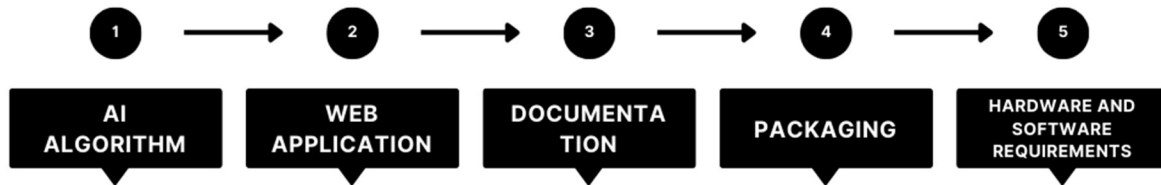
- 1. Higher Accuracy:** By utilizing advanced deep learning architectures and an extensive dataset, we anticipate achieving a level of accuracy that outperforms existing solutions.
- 2. Efficiency:** The system will be optimized for speed, enabling real-time or near-real-time analysis of histopathology images.
- 3. User-Centric Design:** Our web application will prioritize the needs of medical professionals, providing clear and interpretable results.



Final Project Output:

The final project output will be a fully functional AI-based system for IDC detection in breast histopathology images. It will consist of the following components:

- 1. AI Algorithm:** A robust and highly accurate deep learning algorithm, based on state of the art techniques, trained to detect IDC in breast histopathology images.
- 2. Web Application:** A user-friendly web application accessible to medical practitioners. This application will serve as the interface for uploading histopathology images and receiving IDC detection results.
- 3. Documentation:** Comprehensive documentation detailing the system's architecture, usage instructions, and technical specifications.
- 4. Packaging:** The system will be packaged for easy deployment, including any necessary software dependencies.
- 5. Hardware and Software Requirements:** Detailed requirements for the hardware (including GPU specifications) and software (libraries, frameworks, and operating systems) necessary for deploying and running the system will be provided.

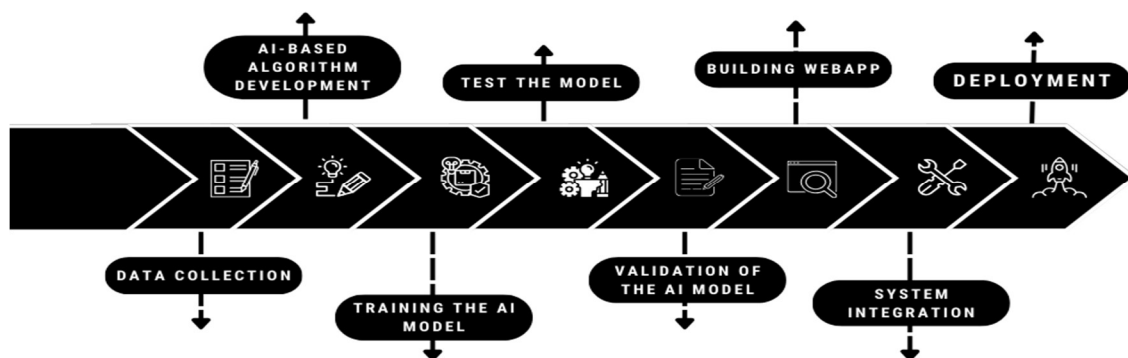


In summary, our project aims to deliver an exceptional AI-based solution for IDC detection in breast histopathology images, setting new standards in terms of accuracy, efficiency, and usability, and potentially advancing the state-of-the-art in breast cancer diagnosis.

Project development methodology

- 1. Data collection:** A large dataset of breast histopathology images will be collected from hospitals. Also online dataset will also be used example kaggle [5], mendeley [6] , paperswithcode [7] etc.
- 2. AI-based algorithm development:** An AI-based algorithm will be developed using state-of-the-art deep learning techniques. The algorithm will be designed to analyze histopathology images and accurately identify the presence of IDC. The algorithm will depend on the accuracy and its efficiency in case of low accuracy and efficiency the algorithm will be change.
- 3. Training the AI model:** The developed AI algorithm will be trained using the collected dataset to improve its accuracy in detecting IDC. The training will be carried out using a high-performance computing platform to accelerate the process. For which we will need GPU because the more the dataset the best the algorithm the higher the resources will be required thus we need GPU.

4. **Test the model:** Testing the model and getting the require result the better the result will smooth the way to its implementation on real life problem.
5. **Validation of the AI model:** The performance of the developed AI model will be evaluated using a test dataset consisting of histopathology images. The evaluation will be carried out by comparing the results of the AI model with the manual interpretation of histopathology images by expert pathologists.
6. **Building WebApp:** The trained model will be then used and a web application will be created for it just because to make it easy for the doctor to access and use it and show the results in better form.
7. **System integration:** The developed AI model will be integrated into a user-friendly software system for use in clinical practice.



Milestone:

1. Project Initiation :

- Define the project scope, objectives, and team roles.
- Gather initial research materials and relevant literature.
- Establish communication channels with healthcare institutions for data collection.

2. Data Collection and Preprocessing :

- Collect a diverse dataset of breast histopathology images.
- Anonymize and preprocess the dataset to remove noise and artifacts.
- Organize the data into training, validation, and test sets.

3. Algorithm Development :

- Develop and fine-tune the AI-based algorithm for IDC detection.
- Experiment with different deep learning architectures and techniques.
- Address any challenges related to model convergence and performance.

4. Model Training :

- Utilize high-performance computing resources, including GPUs, for efficient model training.
- Monitor and optimize the training process to achieve a high level of accuracy.
- Regularly validate the model's performance on a validation dataset.

5. Web Application Development :

- Design and develop a user-friendly web application interface.

- Integrate the trained AI model into the web application.
 - Implement features for uploading histopathology images and displaying results.
- 6. System Testing :**
- Thoroughly test the AI-based system, including functionality and usability.
 - Perform rigorous testing to ensure accurate IDC detection.
 - Address any bugs, issues, or performance bottlenecks.
- 7. Validation and Evaluation :**
- Evaluate the AI model's performance using a separate test dataset.
 - Compare the AI model's results with manual interpretations by expert pathologists.
 - Analyze the system's accuracy, sensitivity, specificity, and precision.
- 8. Documentation and Packaging :**
- Create comprehensive documentation for the AI model, web application, and system architecture.
 - Package the AI system for easy deployment, including software dependencies.

Tasks	September	October	November	December	January
Project Initiation					
Data Collection and Preprocessing					
Algorithm Development					
Model Training					
Web Application Development					
System Testing					
Validation and Evaluation					
Documentation and Packaging					

Future Work

Once the initial project is completed and the AI-based system for IDC detection in breast histopathology images is developed, there are several avenues for future work and improvement:

1. **Enhanced Accuracy:** Continuously improve the accuracy of the AI model by collecting more diverse and extensive datasets, fine-tuning the model architecture, and incorporating advanced machine learning techniques.
2. **Mobile Application:** Creating a Mobile Application so that it can be accessible easily by doctors, from anywhere with ease and this will make the detection faster without losing anytime.
3. **Real-time Processing:** Explore ways to make the AI system capable of real-time image analysis to provide instant results during medical procedures or screenings.
4. **Integration with Healthcare Systems:** Integrate the AI system with existing healthcare databases and electronic health records (EHRs) to streamline the diagnostic process and facilitate seamless communication between AI tools and healthcare providers.

References

- [1] <https://www.who.int/news-room/fact-sheets/detail/breast-cancer>, "World Health Organization," 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/breast-cancer>.
- [2] M. A. T. Q. N. Y. A. O. E. Ali Bou Nassif, "Breast cancer detection using artificial intelligence techniques: A systematic literature review," *www.sciencedirect.com*, 2022.
- [3] M. M. A. A. Md Ishtyaq Mahmud, "A Deep Analysis of Transfer Learning Based Breast Cancer Detection Using Histopathology Images," *arxiv.org*, 2023.
- [4] K. D. L. H. T. M. & T. Q. D. Richard Adam, "Deep learning applications to breast cancer detection by magnetic resonance imaging," *breast-cancer-research.biomedcentral.com*, 2023.
- [5] P. MOONEY, "Breast Histopathology Images," Kaggle.com, [Online]. Available: <https://www.kaggle.com/datasets/paultimothymooney/breast-histopathology-images>.
- [6] H. Bolhasani, "A histopathological image dataset for grading breast invasive ductal carcinomas," *data.mendeley.com*, 2020. [Online]. Available: <https://data.mendeley.com/datasets/w7jcx7gj6/1>.
- [7] L. S. O. C. P. a. L. H. F. A. Spanhol, "BreakHis (Breast Cancer Histopathological Database)," *paperswithcode.com*, [Online]. Available: <https://paperswithcode.com/dataset/breakhis>.

